Literature review over microring modulators.

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- High-speed silicon modulator based on cascaded microring resonators, Y. Hu *et al.* (Optics Express, 2012).
- Silicon microring modulator for 40 Gb/s NRZ-OOK metro networks in O-band (Xuan *et al.*, Optics Express 2014).

- High-speed silicon modulator based on cascaded microring resonators, Y. Hu *et al.* (Optics Express, 2012).

High-speed silicon modulator based on cascaded microring resonators

- Demonstrated a cascaded double microring resonators is, The proposed modulator experimentally achieved **40 Gbit/s** modulation with an extinction ratio of **3.9 dB**.

- Enhancement of the modulator achieves with an ultra-high optical bandwidth of **0.41 nm**, corresponding to **51 GHz**, was accomplished by using cascaded double ring structure.

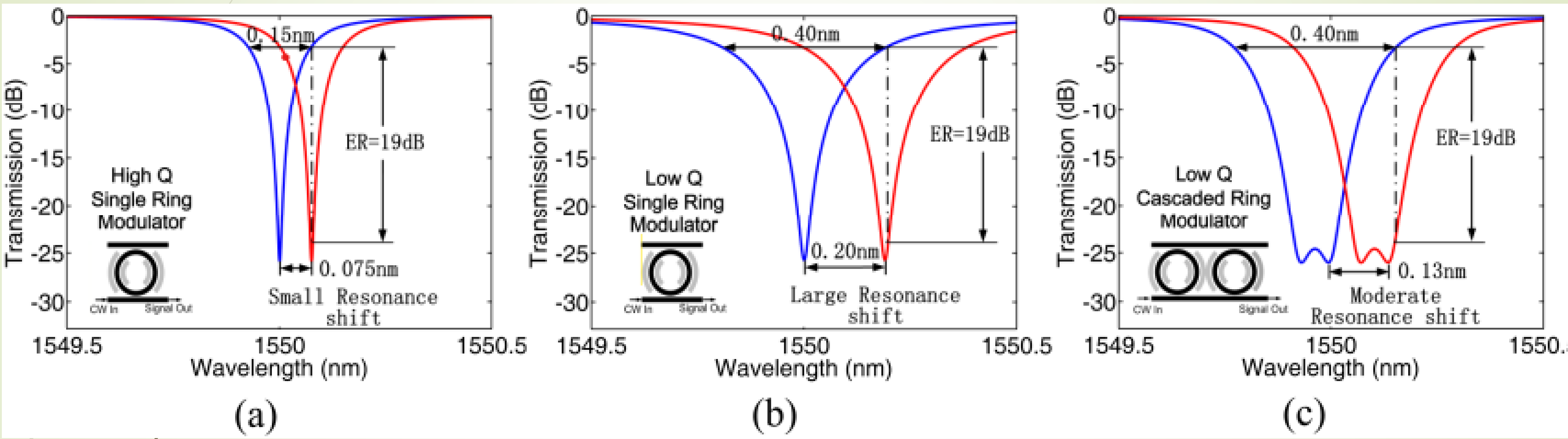
High-speed silicon modulator based on cascaded microring resonators

Why cascaded rings ?

- a sharp resonance (high Q factor) is required to achieve high dynamic extinction ratios with a relatively small resonance shift. However, a high-Q resonator has a longer photon lifetime which limits the achievable modulation bandwidth.

- To relieve the Q-limited problem, coupled-ring-resonator and dual-ring structure are proposed to be used in enhancing the photon-lifetime-induced optical bandwidth.

High-speed silicon modulator based on cascaded microring resonators



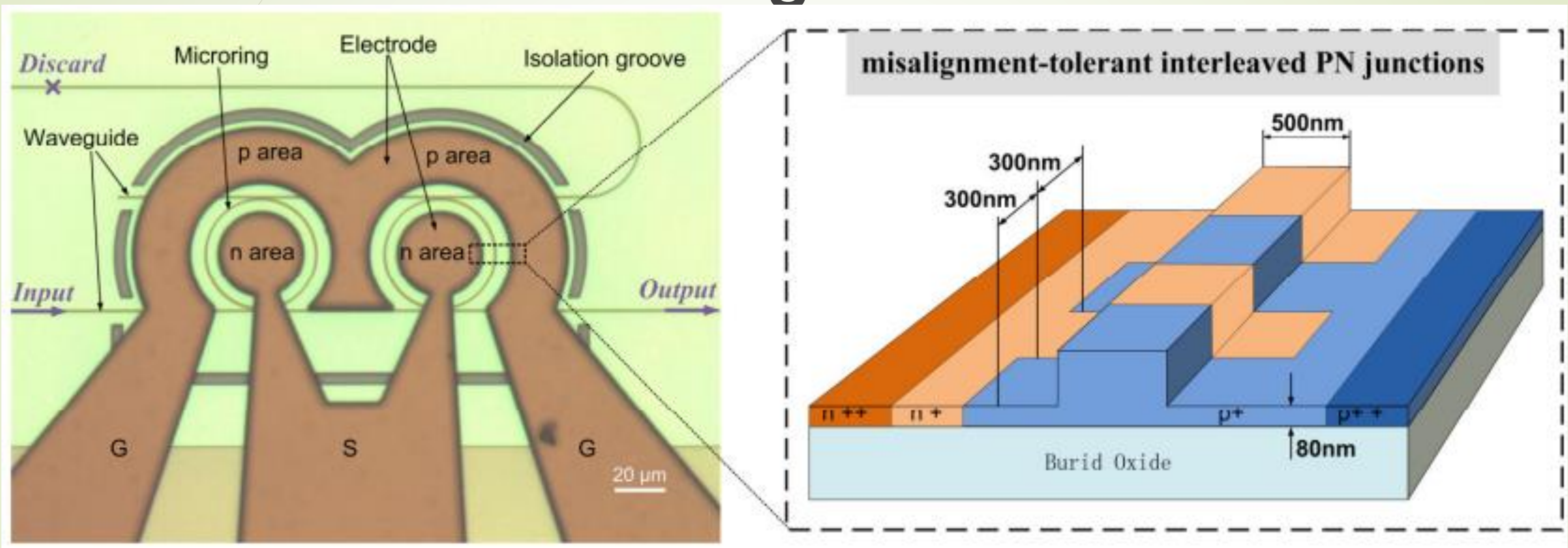
- Fig. 1(a) Single ring modulator scheme with a high Q factor micoring "with a small resonance shift of **0.075 nm** can provide extinction ratio of **19 dB**".
(b) Single ring modulator scheme with a low Q factor micoring.
(c) The proposed cascaded ring modulator scheme where Two rings coupled in series.

High-speed silicon modulator based on cascaded microring resonators

Fabrication & geometry

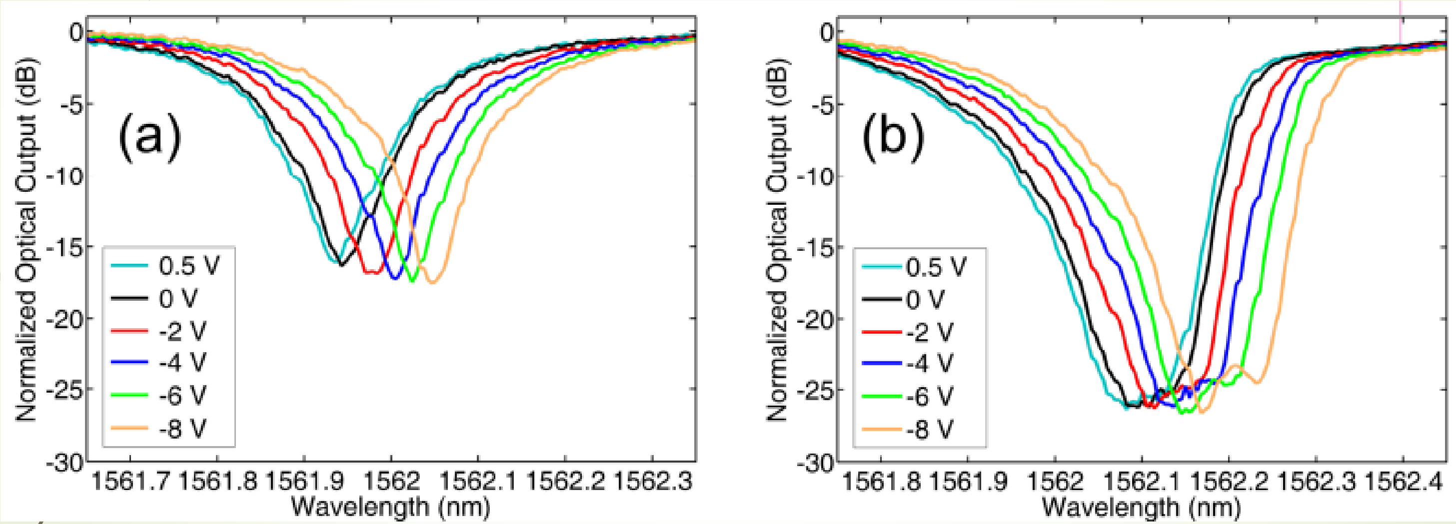
- Ring radius: **20 µm** (each ring identical).
- Waveguides: height **340 nm**, slab **80 nm**; bus width **450 nm**, ring width **500 nm**.
- Optical simulation parameters: effective index n_eff = **2.91**, ring amplitude loss a = **0.995**.
- Coupling coefficients used to simulate the cascaded: t1 = **0.9310**, t2 = **0.9555**.
- Doping for active region: **2×10^{17} cm^{-3}**; contact p++/n++: **1×10^{19} cm^{-3}**.
- The lengths of both p and n region are **300 nm**, making a **600 nm** long doping period.

High-speed silicon modulator based on cascaded microring resonators



- Fig. 2 Top-view of microscopic picture of the fabricated cascaded microring modulator and the schematic 3D view of the interleaved PN junctions.

High-speed silicon modulator based on cascaded microring resonators



- Fig. 3 . Normalized transmission of (a) the fabricated single ring modulator, and (b) cascaded double ring modulator for different applied reverse bias from 0.5 V to – 8 V.

High-speed silicon modulator based on cascaded microring resonators

Eye diagram

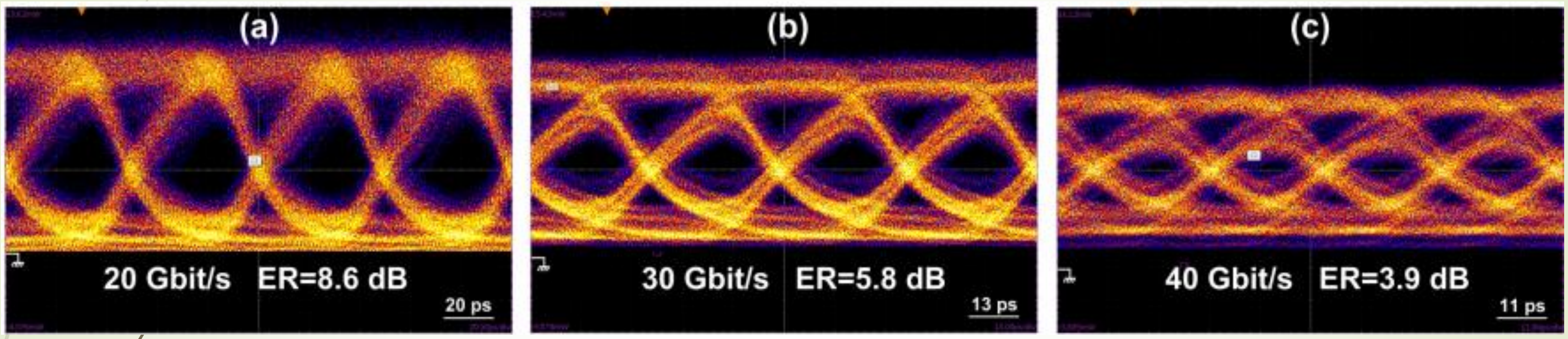


Fig.4 Eye measured at (a) 20 Gbit/s, (b) 30 Gbit/s and (c) 40 Gbit/s eye diagrams of the cascaded ring modulator under The electrical voltage peak-to-peak of 5 V and DC bias of −4.5 V.

- Silicon microring modulator for 40 Gb/s NRZ-OOK metro networks in O-band (Xuan *et al.*, Optics Express 2014).

Silicon microring modulator for 40 Gb/s NRZ, OOK metro networks in O-band

- A microring-based silicon modulator operating at **40 Gb/s** near **1310 nm** is demonstrated for the first time to our knowledge. NRZ-OOK signals at **40 Gb/s** with **6.2 dB** extinction ratio are observed by applying a **4.8** Vpp driving voltage.

- O-band (~**1310 nm**) is near the **minimal-dispersion wavelength of standard SMF**, so high-rate, direct-detected OOK can reach longer metro distances without dispersion compensation (different wavelengths of light travel with different speed).
- At **1550 nm** standard single mode fibers (SSMF) exhibit strong chromatic dispersion (CD), the spreading of optical pulses as they travel through the fiber.

Silicon microring modulator for 40 Gb/s NRZ, OOK metro networks in O-band

Fabrication & geometry

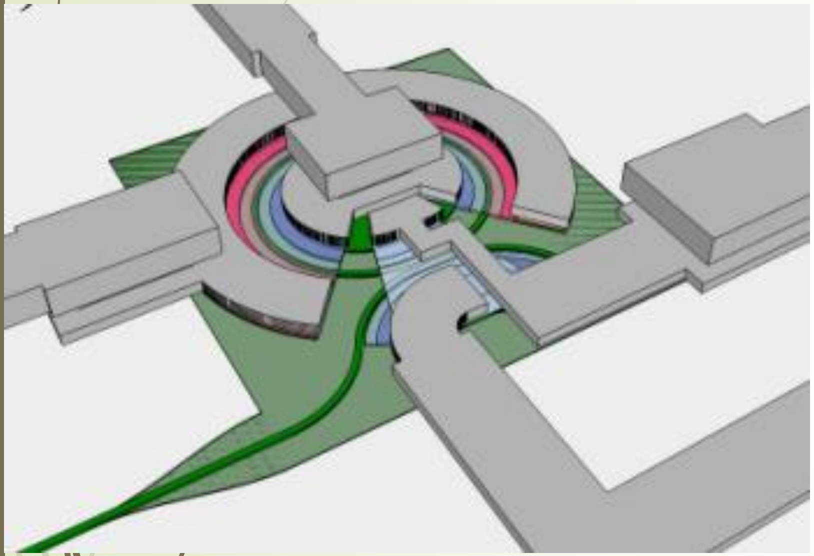


Fig.5 Layout of the modulator. Light green stands for the **90nm** thick Si, dark green for the waveguide, and gray for the back-end metallization. Red stands for **p** doping, and blue for **n** doping. Deeper the color, heavier the dose.

- on an **8 inches** silicon-on-insulator (SOI) wafer, **with 220 nm** thick top silicon, **2 μm** thick buried oxide and high-resistivity (750 Ω·cm) substrate.

Silicon microring modulator for 40 Gb/s NRZ, OOK metro networks in O-band

Fabrication & geometry

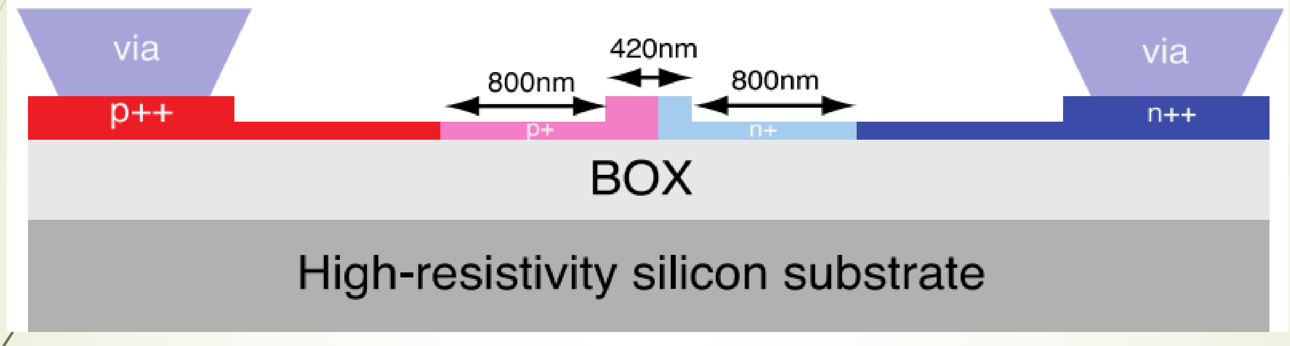


Fig.6 The cross section of the phase shifter

- A lateral p-n junction is formed on the waveguide at doping levels near **2 × 10^18 cm−3**.
- The microring modulator is **15 μm** in diameter, resulting in a free spectral range (FSR) of **9.2 nm**

Silicon microring modulator for 40 Gb/s NRZ, OOK metro networks in O-band

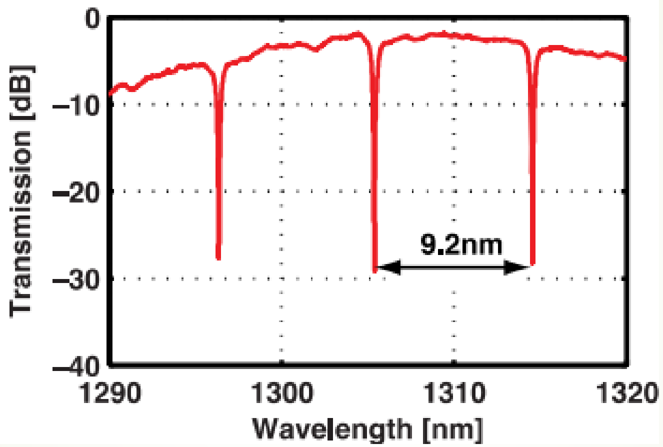


Fig.7 Transmission spectrum of the modulator

- Drive: **4.8 Vpp** electrical drive, DC bias **−2.6 V**, laser tuned to **1314.852 nm** at an insertion-loss point (**≈−7 dB**) to maximize ER.
- The full width at half maximum (FWHM) near 1314 nm **is 0.372** nm

Silicon microring modulator for 40 Gb/s NRZ, OOK metro networks in O-band

Eye diagram

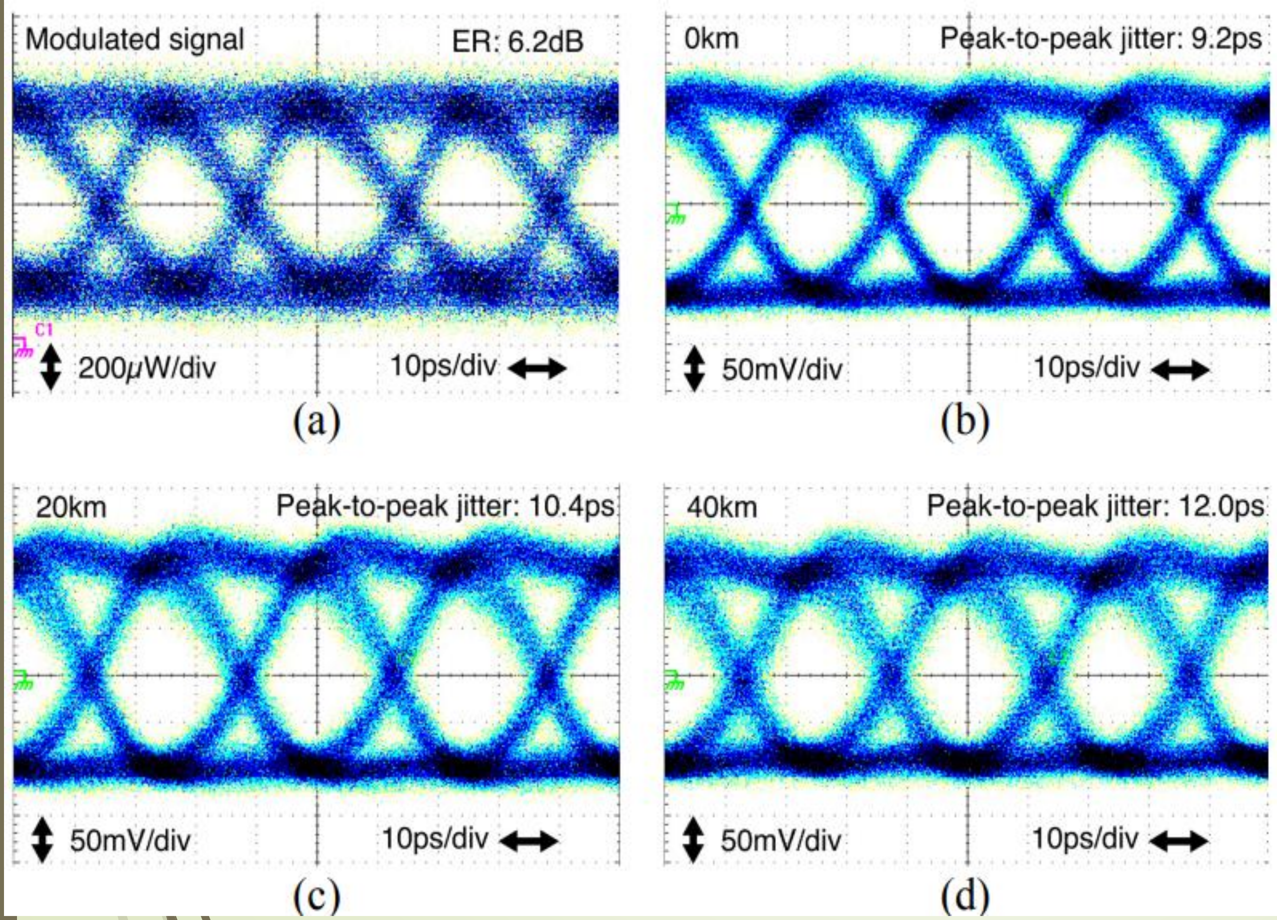


Fig.8 (a) **40 Gb/s** optical eye of the modulated signal. Wavelength: 1314.852 nm, ER: **6.2 dB**.

(b)-(d) **40 Gb/s** electrical eye after 0km/20km/40km transmission.

References

- Hu, Yingtao, et al. "High-speed silicon modulator based on cascaded microring resonators." Optics express 20.14 (2012): 15079-15085.
- Xuan, Zhe, et al. "Silicon microring modulator for 40 Gb/s NRZ-OOK metro networks in O-band." Optics express 22.23 (2014): 28284-28291.